Lab 42 Worksheet — Wireless Security and SOHO Router Hardening

Name: ____ **Date:** __

Exam objective: Configure wireless encryption and harden a SOHO router against the attacks the default configuration invites (Core 2 objectives 2.9 and 2.10).

Record as you go

Step	What you did	What you observed
1	Record the wireless encryption generations in order and their status: WEP is broken and must never be used, WPA with TKIP is deprecated, WPA2 with AES-CCMP is the practical minimum, and WPA3 with SAE is current.	
2	Record what WPA3's SAE handshake adds: it defeats the offline dictionary attack that works against a captured WPA2 handshake.	
3	Change the default administrator password first, and record that default credentials for every common router model are published online.	
4	Change the default SSID to a name that does not identify the router make, model or the household or business, since the model name tells an attacker which exploits to try.	
5	Set WPA2-AES or WPA3 encryption with a passphrase of at least 16 characters, and record why the passphrase length matters against offline cracking.	
6	Record the truth about disabling SSID broadcast: it stops the network appearing in casual scans but any wireless analyser still sees it, so treat it as tidiness rather than security.	
7	Record the truth about MAC filtering: MAC	

Step	What you did	What you observed
	addresses are trivially spoofed once observed, so it deters casual users but stops no capable attacker.	
8	Update the router firmware and record why this matters most of all — router vulnerabilities are publicly disclosed and actively scanned for within days.	
9	Disable WPS, disable remote or WAN-side administration, and disable UPnP unless a specific application requires it, recording the risk each one carries.	
10	Configure the guest network as an isolated SSID with no access to the internal LAN, and verify the isolation from a connected guest device.	
11	Set DHCP reservations for infrastructure devices and record the address plan, verifying the ranges in IP Calculator at https://alfredang.github.io/ipcalculator/.	
12	Complete the hardening checklist ordered by impact, with the threat each step defeats, and mark which three steps you would perform if you had only five minutes.	

Verification

Success criterion: Your checklist covers at least ten hardening steps each with a named threat; WEP, WPA, WPA2 and WPA3 are correctly ordered by security; SSID hiding and MAC filtering are correctly described as weak controls; and firmware update appears in your five-minute priority set.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?